# GridShift

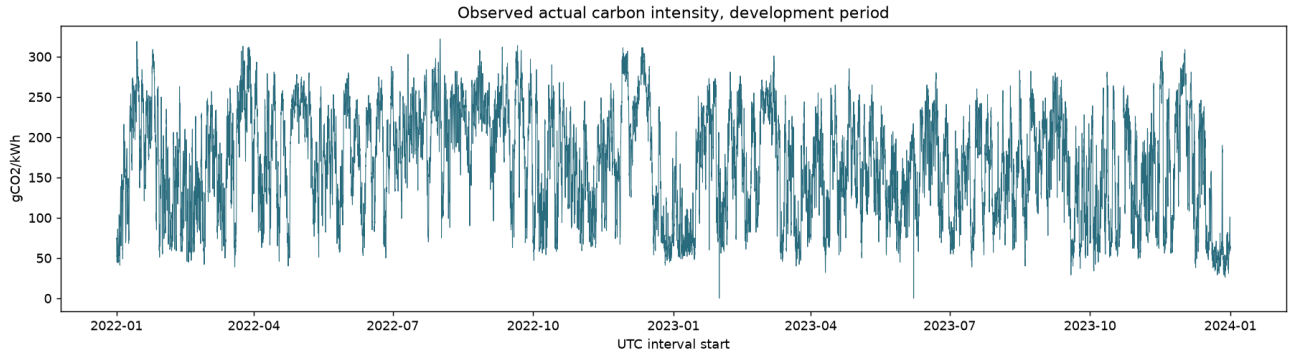
Carbon-aware GPU workload scheduling with classical time-series forecasts

Research report generated from verified local artifacts

Study ID: d2dee2d358735d7d

Target: NESO estimated GB carbon intensity in gCO2/kWh

Scheduling emissions are modeled from declared energy assumptions.



# Abstract

This report evaluates a frozen classical forecasting and scheduling study. It separates validation selection from locked-test measurement.

Modeled isolated-job benefit under declared assumptions. The certified seven-day paired block estimate is 12.414% (95% interval 10.951 to 13.828%) across 2913 scheduled primary-policy jobs; the bootstrap paired 2913 jobs. This estimates modeled operational CO2 under the declared job, energy, and deadline assumptions.

The source series contains NESO estimated actual carbon intensity. The scheduling outcomes are modeled estimates under stated job and energy assumptions.

# Research question and scope

The study asks whether half-hourly forecasts of NESO estimated GB carbon intensity can improve feasible GPU workload placement against earliest-feasible execution and simple seasonal controls.

The primary experiment uses isolated counterfactual jobs. A secondary one-GPU queue replay tests a simple reservation policy under synthetic arrivals.

The carbon oracle observes realized future intensity. It is an unattainable opportunity bound, not a candidate for deployment.

| component | definition |
|---|---|
| Target | NESO estimated actual, gCO2/kWh |
| Forecast path | Half-hourly, configured horizons |
| Primary contrast | Validation-selected schedule versus earliest feasible |
| Queue | One GPU, synthetic arrivals, non-preemptive reservations |

# Data and provenance

The data manifest records normalized hash `f57548d5bc197f884b5b35bb84faaaadcb38b7b46dc3e8c925a150b8ea5a5a1e`.

The configured source is `https://api.carbonintensity.org.uk`. The target spans `2022-01-01T00:00:00Z` through `2026-01-01T00:00:00Z`.

The provider's historical forecast values do not have verified issue-time provenance in the inspected interfaces. They are not treated as fair as-issued competitors. [S1-S5]

# Data-quality audit

The table reproduces the stored quality audit. It reports coverage and grid checks without converting the planned thresholds into measured facts.

The raw reconstruction verified 209 source chunks and 70128 UTC half-hour slots. It identifies 122 intervals without a raw API record and 56 records with an explicit null actual. These categories come from the raw interval audit; diagnostics.input_absent_intervals describes the development-series input and cannot establish raw API absence.

| measure | value |
|---|---|
| slots | 70128 |
| observed | 69950 |
| missing | 178 |
| overall_coverage | 0.9975 |
| minimum_monthly_coverage | 0.9388 |
| minimum_test_monthly_coverage | 1 |
| unique_utc_intervals | true |
| regular_grid | true |
| data_gate_pass | true |
| availability_is_assumed | true |

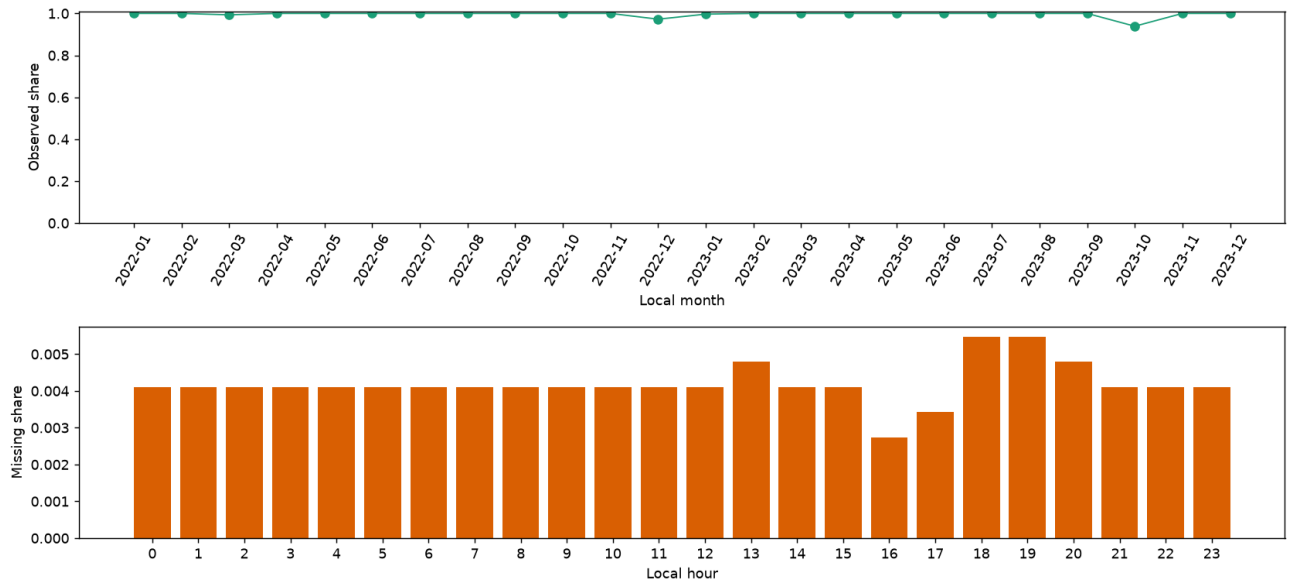



Figure. Development-period missingness by month and local hour. Source: artifacts/diagnostics/figures/missingness.png.

# Development series

This plot uses development observations only. It informs transformations and candidate construction. It is not a test-year result.

The source target remains separate from provider forecasts. Missing target values are not filled from the forecast field. [S2, S4]

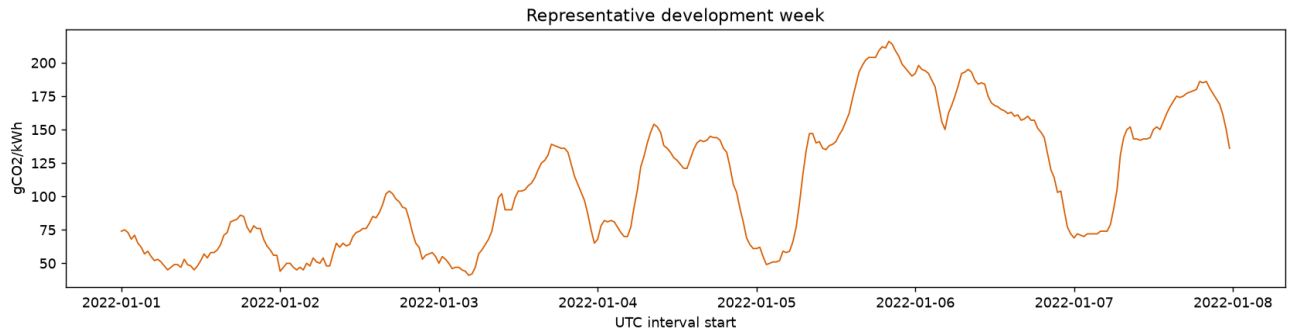



Figure. Representative development week from the stored diagnostics.

# Seasonality and decomposition

The diagnostic record reports daily ACF at lag 48 as `0.52289` and weekly ACF at lag 336 as `0.26307` when those values exist.

Robust STL at period 48 is descriptive. Its centered endpoint behavior cannot support a real-time claim. Residual whiteness does not establish forecast accuracy. [S9-S13]

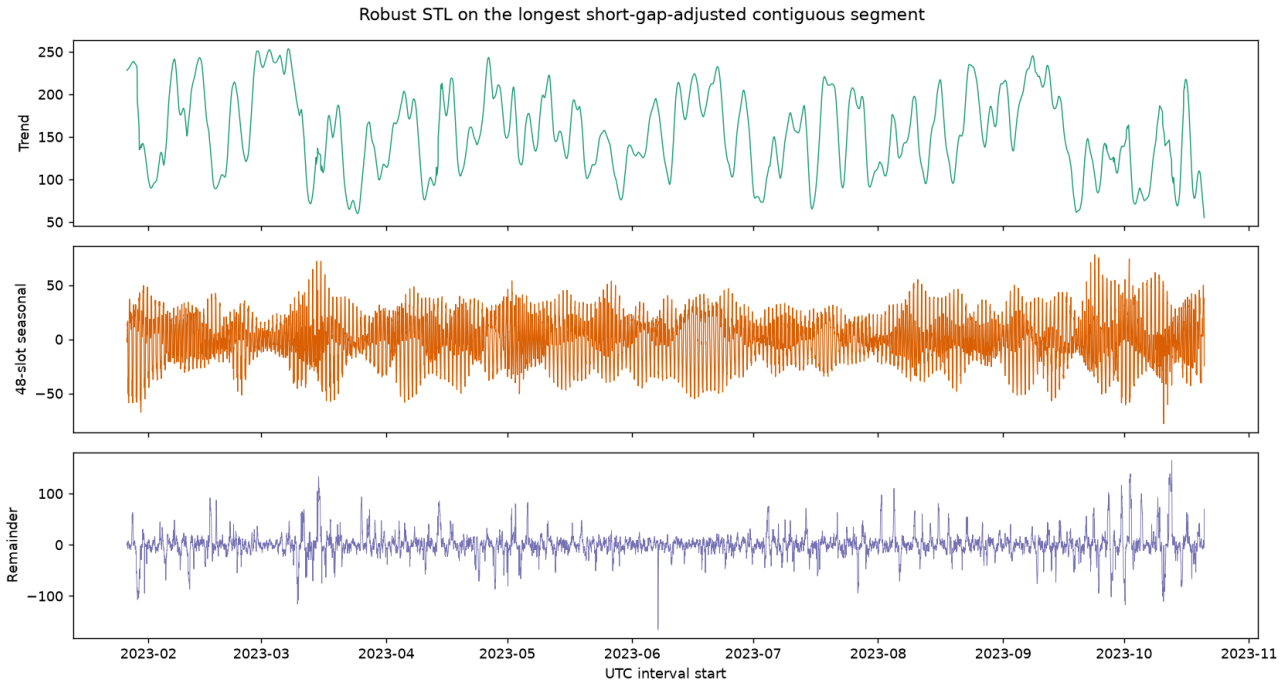



Figure. Robust STL decomposition on the recorded development segment.

# Stationarity diagnostics

ADF tests address a unit-root null under specified deterministic terms. KPSS uses a different null. Conflicting or inconclusive tests do not select a forecasting model by themselves. [S11]

The diagnostics record contains 20 ADF rows and 8 KPSS rows. Missing rows remain absent rather than being filled with assumed p-values.

Recorded level ADF p-values range from 6.769e-21 to 3.5813e-12. Level KPSS p-values are 0.08477, 0.08268. KPSS values at 0.1 or 0.01 are lookup-table bounds, not exact tail probabilities. Read each against its specified regression, sample period, and lag cap. These tests do not establish stability across years or prove white noise.

| variant | regression | autolag | status | sample_n | selected_lag | test_statistic | p_value | critical_5pct |
|---|---|---|---|---|---|---|---|---|
| levels_identity | c | AIC | ok | 12828 | 48 | -11.42 | 6.769e-21 | -2.862 |
| levels_identity | c | BIC | ok | 12828 | 48 | -11.42 | 6.769e-21 | -2.862 |
| levels_identity | ct | AIC | ok | 12828 | 48 | -11.51 | 2.595e-18 | -3.411 |
| levels_identity | ct | BIC | ok | 12828 | 48 | -11.51 | 2.595e-18 | -3.411 |
| levels_log1p | c | AIC | ok | 12828 | 48 | -8.5 | 1.248e-13 | -2.862 |
| levels_log1p | c | BIC | ok | 12828 | 48 | -8.5 | 1.248e-13 | -2.862 |
| levels_log1p | ct | AIC | ok | 12828 | 48 | -8.564 | 3.581e-12 | -3.411 |
| levels_log1p | ct | BIC | ok | 12828 | 48 | -8.564 | 3.581e-12 | -3.411 |
| first_difference_identity | c | AIC | ok | 12827 | 48 | -14.04 | 3.31e-26 | -2.862 |
| first_difference_identity | c | BIC | ok | 12827 | 48 | -14.04 | 3.31e-26 | -2.862 |
| first_difference_log1p | c | AIC | ok | 12827 | 48 | -18.49 | 2.125e-30 | -2.862 |
| first_difference_log1p | c | BIC | ok | 12827 | 48 | -18.49 | 2.125e-30 | -2.862 |

# Forecast model catalogue

The study uses AR, MA, ARMA, ARIMA, and daily-seasonal SARIMA structures. Baselines include persistence, daily and weekly seasonal-naive, and past-only calendar mean.

AR, MA, and ARMA can be nested ARIMA cases. Effective duplicates are counted once in comparisons. Unsuccessful fits remain part of the run ledger. [S9, S10, S12]

AR uses past levels; MA uses past innovations rather than a rolling arithmetic average. ARMA combines those terms. ARIMA applies (1-B)^d before stationary ARMA dynamics. SARIMA adds seasonal polynomials and (1-B^48)^D.

| family | role |
|---|---|
| AR | Stationary autoregression |
| MA | Past innovation dependence |
| ARMA | Stationary mixed dynamics |
| ARIMA | Integrated level model |
| SARIMA | Daily seasonal state-space model |
| Simple baselines | Persistence, seasonal-naive, calendar mean |

# Lag selection and information criteria

AIC = -2 log L + 2k; BIC = -2 log L + k log(n_effective). AICc adds 2k(k+1)/(n_effective-k-1). k includes variance and deterministic parameters.

These representative first-anchor shortlist rows belong to separate family, transformation, and differencing groups. Their absolute IC values must not be pooled into a leaderboard.

All attempted orders and failures remain in selection_ledger.csv. Aggregate within-anchor AICc and BIC ranks shortlist orders; chronological validation chooses representations.

| family | transform | order | aic | bic | aicc | converged_anchors |
|---|---|---|---|---|---|---|
| AR | identity | [7, 0, 0] | 5.4e+04 | 5.407e+04 | 5.4e+04 | 12 |
| MA | identity | [0, 0, 8] | 5.823e+04 | 5.83e+04 | 5.823e+04 | 12 |
| ARMA | identity | [3, 0, 2] | 5.403e+04 | 5.408e+04 | 5.403e+04 | 12 |
| ARIMA | identity | [3, 1, 3] | 5.398e+04 | 5.403e+04 | 5.398e+04 | 12 |
| SARIMA | identity | [1, 0, 1] | 5.462e+04 | 5.465e+04 | 5.462e+04 | 12 |
| AR | log1p | [8, 0, 0] | -2.614e+04 | -2.607e+04 | -2.614e+04 | 12 |
| MA | log1p | [0, 0, 8] | -2.222e+04 | -2.215e+04 | -2.222e+04 | 12 |
| ARMA | log1p | [3, 0, 1] | -2.612e+04 | -2.607e+04 | -2.612e+04 | 12 |
| ARIMA | log1p | [2, 1, 1] | -2.601e+04 | -2.598e+04 | -2.601e+04 | 12 |
| SARIMA | log1p | [1, 0, 1] | -2.572e+04 | -2.569e+04 | -2.572e+04 | 10 |

# Representative validation residual diagnostics

Ljung-Box at lag 48 checks remaining serial dependence with model degrees of freedom. All lags, innovation normality, and ACF/PACF plots remain in validation/residuals/.

| model_id | n_contiguous | lb_stat | lb_pvalue |
|---|---|---|---|
| AR(7,0,0) | 8226 | 1544 | 2.513e-297 |
| MA(0,0,8)[log1p] | 8226 | 1.273e+04 | 0 |
| ARMA(3,0,2) | 8226 | 1203 | 6.107e-224 |
| ARIMA(3,1,3) | 8226 | 1173 | 1.653e-218 |
| SARIMA(1,0,1)x(1,0,0,48) | 8226 | 2459 | 0 |

Small p-values indicate remaining serial dependence. The innovation-distribution table accompanies the report. Printed zero p-values represent numerical underflow. These diagnostics and imperfect predictive coverage prevent a calibrated Gaussian-uncertainty claim.

# Chronological evaluation design

Forecasts use eligible history, persist before scoring, and are compared on common target masks. The test year remains outside validation selection.

Report MAE, RMSE, mean error, daily-seasonal MASE, predictive interval coverage, interval width, and sample counts by lead time. Avoid MAPE when targets may be zero. [S9-S13]

| setting | value |
|---|---|
| Validation start | 2024-01-01T00:00:00Z |
| Validation end | 2025-01-01T00:00:00Z |
| Test start | 2025-01-01T00:00:00Z |
| Test end | 2026-01-01T00:00:00Z |
| Reported horizons | [1, 4, 12, 24, 48] |
| Forecast path slots | 48 |

# Validation forecast outcomes

The validation-selected forecast champion is `SARIMA(1,0,1)x(1,0,0,48)`.

The strongest simple validation baseline is `persistence`. These selections precede locked-test scoring.

These are common-origin scores. End-to-end scores and all required families remain in the accompanying Markdown report and source tables.

| model_id | group | horizon | n | mae | rmse | mean_error | mase | coverage80 | coverage95 |
|---|---|---|---|---|---|---|---|---|---|
| SARIMA(1,0,1)x(1,0,0,48) | all | 1 | 15683 | 7.124 | 9.864 | -0.02633 | 0.1736 | 0.8325 | 0.9371 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 4 | 15680 | 15.05 | 20.58 | -0.05673 | 0.3671 | 0.7511 | 0.8854 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 12 | 15672 | 27.92 | 36 | -0.1264 | 0.6808 | 0.6681 | 0.8422 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 24 | 15660 | 33.82 | 43.57 | -0.2568 | 0.8226 | 0.6982 | 0.872 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 48 | 15654 | 40.21 | 51.23 | -0.3095 | 0.9783 | 0.7198 | 0.8885 |
| persistence | all | 1 | 15683 | 8.781 | 12.45 | -0.02002 | 0.2146 | 0.9996 | 1 |
| persistence | all | 4 | 15680 | 18.99 | 26.11 | -0.03705 | 0.4647 | 0.9758 | 0.9986 |
| persistence | all | 12 | 15672 | 35.44 | 45.41 | -0.08518 | 0.868 | 0.8613 | 0.9731 |
| persistence | all | 24 | 15660 | 39.12 | 50.36 | -0.2317 | 0.9565 | 0.8341 | 0.9545 |
| persistence | all | 48 | 15654 | 41.43 | 54.26 | -0.4704 | 1.01 | 0.8085 | 0.9367 |

# Locked-test forecast outcomes

These rows report the recorded locked-test results for the validation-selected champion and strongest simple baseline.

The exporter does not choose a new winner from test outcomes. A post-test revision requires another untouched evaluation period.

| model_id | group | horizon | n | mae | rmse | mean_error | mase | coverage80 | coverage95 |
|---|---|---|---|---|---|---|---|---|---|
| SARIMA(1,0,1)x(1,0,0,48) | all | 1 | 16465 | 6.926 | 9.362 | -0.02102 | 0.1766 | 0.814 | 0.9309 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 4 | 16465 | 14.88 | 20.08 | -0.05526 | 0.3788 | 0.726 | 0.8695 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 12 | 16465 | 27.79 | 35.85 | -0.1243 | 0.7074 | 0.637 | 0.8212 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 24 | 16465 | 33.41 | 43.22 | -0.1281 | 0.8524 | 0.6895 | 0.8508 |
| SARIMA(1,0,1)x(1,0,0,48) | all | 48 | 16465 | 39.26 | 50.73 | 0.1809 | 1.002 | 0.7063 | 0.87 |
| persistence | all | 1 | 16465 | 9.101 | 12.57 | -0.04422 | 0.2313 | 0.9999 | 1 |
| persistence | all | 4 | 16465 | 20.07 | 27.17 | -0.1076 | 0.5095 | 0.9719 | 0.9991 |
| persistence | all | 12 | 16465 | 37.39 | 47.63 | -0.2743 | 0.9491 | 0.8256 | 0.9636 |
| persistence | all | 24 | 16465 | 39.25 | 50.55 | -0.4479 | 1 | 0.8039 | 0.9446 |
| persistence | all | 48 | 16465 | 39.8 | 51.8 | -0.4904 | 1.017 | 0.8012 | 0.9288 |

# Paired forecast-loss comparison

Positive loss reduction favors the validation-selected classical model over the validation-selected simple forecast. The equal-horizon common-origin contrast is secondary and descriptive.

| block_days | paired_origin_count | estimate_loss_reduction_pct | ci95_loss_reduction_pct_low | ci95_loss_reduction_pct_high |
|---|---|---|---|---|
| 7 | 16465 | 16.03 | 13.37 | 18.7 |

# Cumulative forecast-path errors

Path MAE and RMSE score every half-hour in each prefix in gCO2/kWh. Prefix-mean MAE scores the error of the time-average intensity.

Only complete observed prefixes are scored. Common-origin masks exclude any competitor fallback. These descriptive results do not change endpoint-based selection.

| model_id | horizon | n_complete_paths | path_mae | path_rmse | prefix_mean_mae |
|---|---|---|---|---|---|
| SARIMA(1,0,1)x(1,0,0, 48) | 1 | 16465 | 6.926 | 9.362 | 6.926 |
| SARIMA(1,0,1)x(1,0,0, 48) | 4 | 16465 | 11 | 15.42 | 10.63 |
| SARIMA(1,0,1)x(1,0,0, 48) | 12 | 16465 | 19.04 | 26.45 | 17.59 |
| SARIMA(1,0,1)x(1,0,0, 48) | 24 | 16465 | 25.14 | 34.08 | 21.71 |
| SARIMA(1,0,1)x(1,0,0, 48) | 48 | 16465 | 31.43 | 42.08 | 25.54 |
| persistence | 1 | 16465 | 9.101 | 12.57 | 9.101 |
| persistence | 4 | 16465 | 14.69 | 20.81 | 14.38 |
| persistence | 12 | 16465 | 25.83 | 35.65 | 23.99 |
| persistence | 24 | 16465 | 32.23 | 43.08 | 27.37 |
| persistence | 48 | 16465 | 37.54 | 49.57 | 30.31 |

# Scheduling formulation

For a candidate start, the scheduler sums forecast intensity times interval-overlap energy across the full job duration. It obeys hard deadlines and capacity. Exact cost ties favor the earliest start.

Primary releases are independent counterfactual cases. They are not a simultaneous workload trace. The one-GPU queue simulation uses synthetic arrivals and immutable reservations. [S5]

| assumption | value |
|---|---|
| Default duration (hours) | 2 |
| Default slack (hours) | 10 |
| Declared power (kW) | 0.4 |
| PUE | 1.2 |
| Primary selected model | calendar_mean |

# Primary scheduling outcomes

Modeled isolated-job benefit under declared assumptions. The certified seven-day paired block estimate is 12.414% (95% interval 10.951 to 13.828%) across 2913 scheduled primary-policy jobs; the bootstrap paired 2913 jobs. This estimates modeled operational CO2 under the declared job, energy, and deadline assumptions.

A simple-baseline deployment choice does not establish added scheduling value from classical forecasting. Persistence's flat path resolves to earliest execution. Read all policy rows before attributing the benefit to model complexity.

| policy | model_id | n_scheduled | mean_realized_grams | paired_savings_pct | oracle_regret_grams |
|---|---|---|---|---|---|
| earliest | NA | 2913 | 124 | NA | 26.81 |
| seeded_random | NA | 2913 | 123.5 | 0.4067 | 26.3 |
| oracle | oracle | 2913 | 97.22 | 21.61 | 0 |
| forecast | AR(7,0,0) | 2913 | 114.1 | 8.017 | 16.87 |
| forecast | ARIMA(3,1,3) | 2913 | 114.8 | 7.477 | 17.53 |
| forecast | ARMA(3,0,2) | 2913 | 113.9 | 8.198 | 16.64 |
| forecast | MA(0,0,8)[log1p] | 2913 | 118.3 | 4.639 | 21.05 |
| forecast | SARIMA(1,0,1)x(1,0,0,48) | 2913 | 105.7 | 14.81 | 8.436 |
| forecast | calendar_mean | 2913 | 108.6 | 12.41 | 11.41 |
| forecast | daily_naive | 2913 | 107.7 | 13.14 | 10.51 |
| forecast | persistence | 2913 | 124 | 0 | 26.81 |
| forecast | weekly_naive | 2913 | 107.6 | 13.25 | 10.37 |

# Primary uncertainty

The paired moving-block bootstrap keeps neighboring cases together. The seven-day estimate is primary when present; three- and fourteen-day estimates are sensitivity checks.

A non-positive interval or a null estimate remains a valid negative result. A positive estimate does not establish measured or causal grid emissions.

| block_days | paired_job_count | estimate_savings_pct | ci95_savings_pct_low | ci95_savings_pct_high | is_primary_block |
|---|---|---|---|---|---|
| 3 | 2913 | 12.41 | 11.1 | 13.65 | False |
| 7 | 2913 | 12.41 | 10.95 | 13.83 | True |
| 14 | 2913 | 12.41 | 10.64 | 14.07 | False |

# Selected policy versus the strongest simple forecast

No classical deployment was selected; selected-policy improvement is simple timing value, not classical added value

| paired_job_count | estimate_savings_pct | ci95_savings_pct_low | ci95_savings_pct_high |
|---|---|---|---|
| 2913 | 12.41 | 10.95 | 13.83 |

All declared duration/slack scenarios are reproduced in the complete scenario table in gridshift_report.md and the primary_summary.csv source. They remain secondary descriptive comparisons.

# Monthly scheduling outcomes

Monthly figures show the default duration and slack where the source table records both. They describe temporal variation and do not replace the prespecified aggregate contrast.

| month_utc | scenario_id | policy | model_id | n_eligible_jobs | n_scheduled | mean_realized_grams | paired_savings_pct | fallback_rate |
|---|---|---|---|---|---|---|---|---|
| 2025-01 | duration_2h_slack_10h | earliest | NA | 248 | 248 | 161.5 | NA | 0 |
| 2025-01 | duration_2h_slack_10h | seeded_random | NA | 248 | 248 | 162.2 | -0.4032 | 0 |
| 2025-01 | duration_2h_slack_10h | oracle | oracle | 248 | 248 | 135.3 | 16.23 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | AR(7,0,0) | 248 | 248 | 151.4 | 6.252 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | ARIMA(3,1,3) | 248 | 248 | 153.1 | 5.203 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | ARMA(3,0,2) | 248 | 248 | 151.7 | 6.106 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | MA(0,0,8)[log1p] | 248 | 248 | 157.5 | 2.487 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | SARIMA(1,0,1)x(1,0,0,48) | 248 | 248 | 142.8 | 11.62 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | calendar_mean | 248 | 248 | 140.1 | 13.27 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | daily_naive | 248 | 248 | 145.6 | 9.862 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | persistence | 248 | 248 | 161.5 | 0 | 0 |
| 2025-01 | duration_2h_slack_10h | forecast | weekly_naive | 248 | 248 | 149.1 | 7.679 | 0 |
| 2025-02 | duration_2h_slack_10h | earliest | NA | 224 | 224 | 141.8 | NA | 0 |
| 2025-02 | duration_2h_slack_10h | seeded_random | NA | 224 | 224 | 140.1 | 1.171 | 0 |

# Queue replay outcomes

Queue emissions must be read with offered jobs, completed jobs, rejections, deadline misses, and unmatched work. A lower total caused by rejected work is not a carbon saving.

The queue summary comes from the `test` artifact path.

| offered_load | seed | slack_hours | policy | offered_jobs | completed_jobs | rejected_jobs | matched_savings_pct | interpretation |
|---|---|---|---|---|---|---|---|---|
| 0.3 | 1729 | 6 | earliest | 1411 | 1404 | 7 | 0 | descriptive completion intersection; service differs |
| 0.3 | 1729 | 6 | selected | 1411 | 1380 | 31 | 7.322 | descriptive completion intersection; service differs |
| 0.3 | 1729 | 6 | strongest_simple | 1411 | 1404 | 7 | 0 | descriptive completion intersection; service differs |
| 0.3 | 1729 | 12 | earliest | 1411 | 1410 | 1 | 0 | descriptive completion intersection; service differs |
| 0.3 | 1729 | 12 | selected | 1411 | 1404 | 7 | 9.207 | descriptive completion intersection; service differs |
| 0.3 | 1729 | 12 | strongest_simple | 1411 | 1410 | 1 | 0 | descriptive completion intersection; service differs |
| 0.3 | 2718 | 6 | earliest | 1324 | 1317 | 7 | 0 | descriptive completion intersection; service differs |
| 0.3 | 2718 | 6 | selected | 1324 | 1294 | 30 | 5.703 | descriptive completion intersection; service differs |
| 0.3 | 2718 | 6 | strongest_simple | 1324 | 1317 | 7 | 0 | descriptive completion intersection; service differs |
| 0.3 | 2718 | 12 | earliest | 1324 | 1324 | 0 | 0 | descriptive completion intersection; service differs |
| 0.3 | 2718 | 12 | selected | 1324 | 1318 | 6 | 8.291 | descriptive completion intersection; service differs |
| 0.3 | 2718 | 12 | strongest_simple | 1324 | 1324 | 0 | 0 | descriptive completion intersection; service differs |

# Bounded sensitivity checks

Six forecasting variants use four three-day quarterly episodes in validation year 2024. These checks do not establish full-year robustness.

Constant power and PUE scale absolute costs algebraically. They cannot establish robustness to measured energy variation.

| month | variant | n_scheduled | paired_savings_pct |
|---|---|---|---|
| 1 | default | 17 | 18.07 |
| 1 | delay_0min | 17 | 18 |
| 1 | delay_60min | 17 | 18.14 |
| 1 | window_90days | 17 | 18.08 |
| 1 | window_365days | 17 | 18.07 |
| 1 | native_missing | 17 | 18.07 |
| 4 | default | 17 | 2.424 |
| 4 | delay_0min | 17 | 2.694 |
| 4 | delay_60min | 17 | 3.246 |
| 4 | window_90days | 17 | 2.424 |
| 4 | window_365days | 17 | 5.926 |
| 4 | native_missing | 17 | 2.424 |

# Deployment and demonstration

The study records `calendar_mean` as the validation-selected scheduling model.

This export does not claim a live GPU scheduler, Slurm integration, cloud workload execution, or public deployment. A local dashboard demo must use the saved versioned artifacts and preserve stale, missing, and infeasible states.

At an illustrative supply intensity of 100 gCO2/kWh, gross savings permit 0.15397 kWh of overhead per isolated counterfactual case. Computing energy is unmeasured, so no net-emissions claim follows.

# Limitations

The NESO series is estimated generation-related CO2. It is not a complete lifecycle or all-GHG inventory. [S2, S5]

Historical publication latency and revisions remain uncertain. Without verified vintages, retrospective replay is pseudo-real-time. [S2, S4]

Power, PUE, duration, and synthetic arrival assumptions do not establish measured GPU behavior.

Retrospective NESO estimated actuals have no verified historical publication vintages.

Thirty-minute publication delay is an assumed replay policy.

Power, PUE, known durations, and arrivals are synthetic. No GPU energy was measured.

Costs are modeled intensity-weighted operational CO2, not actual avoided grid emissions.

National GB intensity is a proxy for a hypothetical GB-powered workload.

Predictive intervals condition on fitted parameters and may be poorly calibrated.

Queue completion intersections are descriptive when policies accept different jobs.

Net computing overhead, marginal emissions, and embodied emissions are unmeasured.

The untouched test covers 2025 only. No post-test model promotion is justified.

Local packaging and runtime verification do not establish public deployment.

Modeled isolated-job benefit under declared assumptions. The certified seven-day paired block estimate is 12.414% (95% interval 10.951 to 13.828%) across 2913 scheduled primary-policy jobs; the bootstrap paired 2913 jobs. This estimates modeled operational CO2 under the declared job, energy, and deadline assumptions.

# Reproducibility and evidence map

Results manifest SHA-256: 8cd196dbbe13512f1e09f2c07e285ce4f476bae673184ab8cd56b92fae58d743.

Data hash: f57548d5bc197f884b5b35bb84faaaadcb38b7b46dc3e8c925a150b8ea5a5a1e.

Configuration digest: 6f0831112aaed65643e9f88b6d31ac5d31222897d4484c461d9103ac94810ccb.

Protocol hash: b6876f7c8f058f9242eb5852a580384bfa0a1365c76f69a509fa975f8377c68c.

Software source hash: 16d9038d1d153c6d68293c5cc8ef0a902381366d5b8ecd993c9cccf03477fda0.

Study ID: d2dee2d358735d7d.

| claim | evidence |
| --- | --- |
| Validation selection | results.json selections and the recorded order-search ledger when present |
| Locked-test forecast result | test/forecast_metrics.csv |
| Primary scheduling estimate | C:\Users\viraj\Documents\Codex\2026-10-22\plan-gridshift-a-complete-ats-mini\outputs\GridShift\artifacts\runs\d2dee2d358735d7d\test\single_job\primary_summary.csv |
| Paired uncertainty | C:\Users\viraj\Documents\Codex\2026-10-22\plan-gridshift-a-complete-ats-mini\outputs\GridShift\artifacts\runs\d2dee2d358735d7d\test\single_job\primary_bootstrap.csv |
| Queue service and emissions | C:\Users\viraj\Documents\Codex\2026-10-22\plan-gridshift-a-complete-ats-mini\outputs\GridShift\artifacts\runs\d2dee2d358735d7d\test\queue\queue_summary.csv |

# References

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[S2] Official API definitions. https://carbon-intensity.github.io/api-definitions/

[S3] NESO national carbon-intensity dataset. https://www.neso.energy/data-portal/national-carbon-intensity-forecast

[S4] NESO national resource dictionary. https://www.neso.energy/data-portal/national-carbon-intensity-forecast/national_carbon_intensity_forecast

[S5] NESO national carbon-intensity methodology. https://api.neso.energy/dataset/f406810a-1a36-48d2-b542-1dfb1348096e/resource/36638178-e993-4d34-afdc-4ada83993585/download/neso-ci-national-methodology_v2.pdf

[S6] Carbon Intensity API terms. https://github.com/carbon-intensity/terms

[S7] NESO Open Data Licence. https://www.neso.energy/data-portal/neso-open-licence

[S8] NESO historic generation mix and carbon intensity. https://www.neso.energy/data-portal/historic-generation-mix

[S9] Forecasting: Principles and Practice, forecasting toolbox. https://otexts.com/fpp3/toolbox.html

[S10] Forecasting: Principles and Practice, ARIMA. https://otexts.com/fpp3/arima.html

[S11] statsmodels augmented Dickey-Fuller test. https://www.statsmodels.org/stable/generated/statsmodels.tsa.stattools.adfuller.html

[S12] statsmodels SARIMAX. https://www.statsmodels.org/stable/generated/statsmodels.tsa.statespace.sarimax.SARIMAX.html

[S13] statsmodels SARIMAX missing-data example. https://www.statsmodels.org/stable/examples/notebooks/generated/statespace_sarimax_internet.html